Current–Voltage Characterization and Parameter Extraction of Diodes and LEDs

Izaiah Thigpen

Diode I–V Characterization and Shockley-Model Parameter Extraction

Abstract

Forward current–voltage (I–V) sweeps were measured for five semiconductor devices — a silicon diode, a 1N5234 Zener, and red, yellow, and green LEDs — and analyzed with the Shockley diode equation to extract the ideality factor (n) and saturation current (I_s). Log-linear fits of the exponential region gave $R^2 > 0.98$ for every device. The silicon junctions were near-ideal (n ≈ 1.1–1.3) with ~0.75 V turn-on, whereas the LEDs exhibited higher ideality factors (n ≈ 1.7–2.4) and turn-on voltages of ~1.8–2.0 V, consistent with recombination-dominated conduction in their wider-bandgap junctions.

Experimental Method

Each device was swept in forward bias while recording terminal voltage and current. The measured data were reduced with a Python/NumPy script that isolates the exponential conduction region (1 μA ≤ I ≤ 5 mA, avoiding the low-current noise floor and the high-current series-resistance roll-off) and performs a linear least-squares fit of ln(I) versus V. From the fit slope and intercept, the ideality factor $n = 1/(\text{slope}\cdot V_T)$ and saturation current $I_s = \exp(\text{intercept})$ were obtained, with $V_T = kT/q \approx 25.9$ mV at room temperature; turn-on voltage was defined at a 1 mA forward current.

Results

Device	n	I_s (A)	V_{on} @1 mA (V)	R^2
Silicon diode	1.11	2.7e-15	0.76	0.9998
1N5234 (Zener, fwd)	1.30	2.0e-13	0.75	0.9994
Red LED	1.70	1.9e-21	1.79	0.9973
Yellow LED	2.44	3.8e-17	1.96	0.9916
Green LED	1.66	7.0e-23	1.88	0.9887

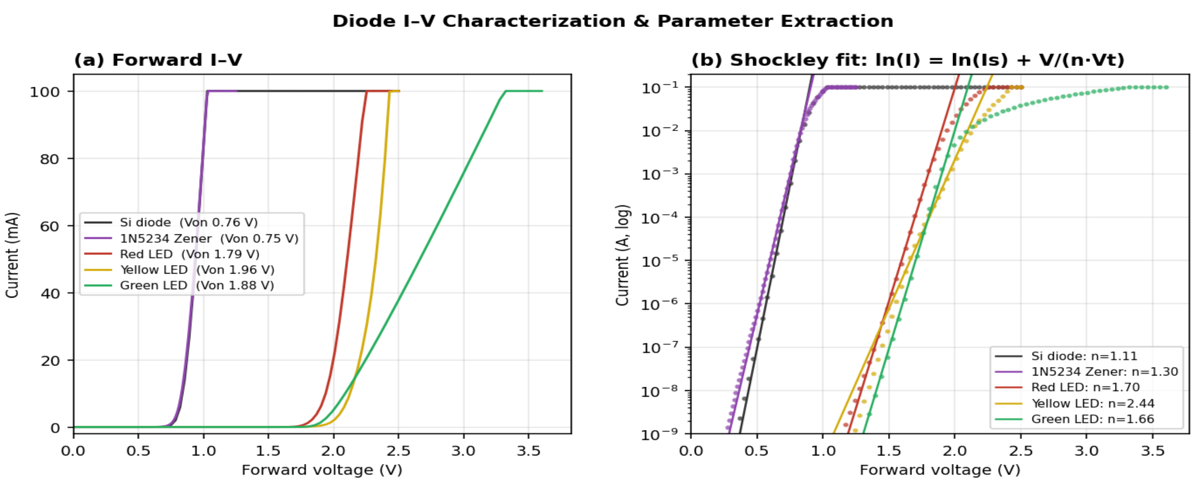

Figure 1. (a) Linear forward I–V showing the higher turn-on of the LEDs; (b) semilog I–V with Shockley fits used to extract n and I_s.

Discussion

Silicon devices. The silicon diode and Zener returned ideality factors near unity (n = 1.11 and 1.30) and ~0.75 V turn-on, in agreement with ideal diffusion-dominated p–n junction behavior and the expected silicon forward drop.

LEDs. The LEDs showed larger ideality factors (n = 1.70–2.44), characteristic of the increased carrier recombination in direct-bandgap III–V materials, and turn-on voltages of ~1.8–2.0 V. The elevated turn-on is consistent with the larger bandgap required for visible emission, and the extremely small extracted saturation currents follow the expected inverse dependence of I_s on bandgap. High-current curvature in the semilog data reflects series resistance, which bounds the useful fitting range.

Conclusion

Shockley-equation fits ($R^2 > 0.98$) cleanly separated near-ideal silicon junctions (n ≈ 1.1–1.3, ~0.75 V) from the higher-ideality, higher-turn-on LEDs (n ≈ 1.7–2.4, ~1.8–2.0 V), demonstrating diode parameter extraction and the link between junction turn-on and semiconductor bandgap.